SOGO: A Scaling Ontology for Geospatial Observations in Knowledge Graphs

Amin Anjomshoaa

Vienna University of Economics and Business (WU)
Institute for Data, Process and Knowledge Management



www.wu.ac.at/dpkm



Introduction

- Scale is a foundational concept in the geographical sciences systems
- Spatial scaling refers to how the distribution of individuals, resources, or phenomena varies across different spatial extents or resolutions.
- The first scientific principle of spatial scaling, known as the square–cube law, was introduced by Galileo Galilei in his 1638 work “Two New Sciences”.
- Galileo examined how changes in an object's size affect its geometric and physical properties, particularly the relationships between its surface area, volume, and weight.


DISCORSI
E
DIMOSTRAZIONI
MATEMATICHE,
intorno à due nuoue scienze
Attenenti alla
MECANICA & i MOVIMENTI LOCALI,
del Signor
GALILEO GALILEI LINCEO,
Filosofo e Matematico primario del Serenissimo
Grand Duca di Toscana.
Con una Appendice del centro di gravità d'alcuni Solidi.
IN LEIDA,
Appresso gli Elsevirii. M. D. C. XXXVIII.

https://en.wikipedia.org/wiki/Square%E2%80%93cube_law

Scaling in Knowledge Graphs

- Spatio-Temporal Knowledge Graphs (STKGs) are widely used in applications such as environmental monitoring, urban planning, and logistics.
- Data is typically collected to meet specific project requirements, while the burden of scaling is shifted to data consumers who want to reuse it.
- Despite extensive research in statistical and spatial analysis, the semantic representation of observation scaling in knowledge graphs is largely unexplored.
- Goal of this research is to model and implement the scale concept as a first-class citizen of knowledge graphs.

Grid-Based Scaling - H3 Example



Upscaling

Downscaling



Political-Border-Based Scaling

#	Stemma	Comune	Provincia	Popolazione (ab)	Superficie (km²)	Densità (ab./km²)	Altitudine (m s.l.m.)
1		Bari	Città metropolitana di Bari	316 245	117,39	2 736,37	5
2		Taranto	Provincia di Taranto	189 355	249,86[104]	798,69	15
3		Foggia	Provincia di Foggia	146 563	509,26	296,73	76
4		Andria	Provincia di Barletta-Andria-Trani	97 192	402,89	247,38	151
5		Lecce	Provincia di Lecce	95 273	238,93	399,06	49
6		Barletta	Provincia di Barletta-Andria-Trani	92 515	149,35	633,05	15
7		Brindisi	Provincia di Brindisi	83 269	332,98	260,71	15
8		**Altamura**	Città metropolitana di Bari	70 355	431,38	164,07	467



https://it.wikipedia.org/wiki/Puglia

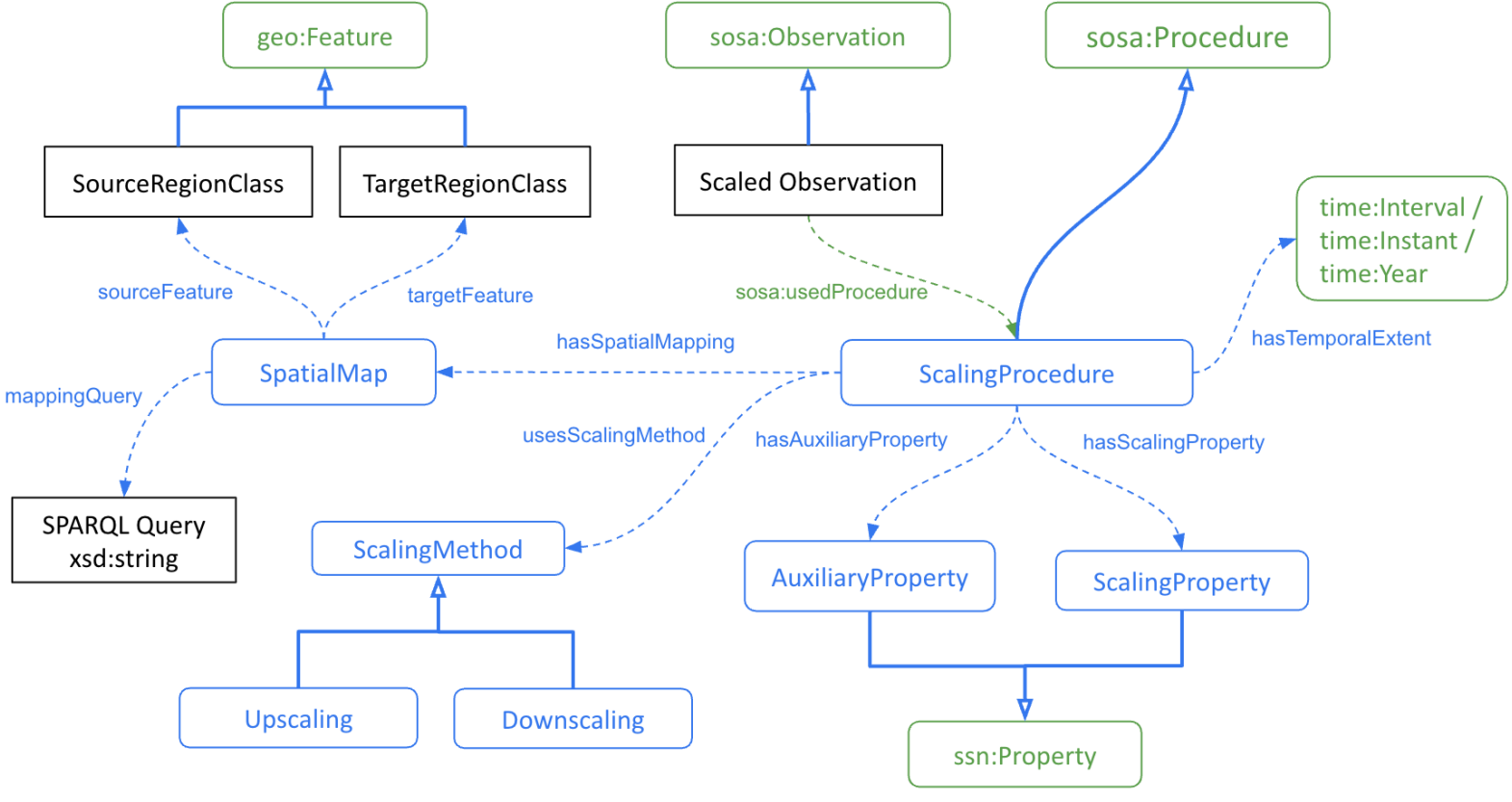
Scaling Procedures

- When scaling observations across spatial sub-regions, there is no single universal formula
- The appropriate scaling approach depends primarily on the nature of the property being represented.
  - **Intensive observations** do not depend on the size of the area and are generally aggregated using averages or other measures.
    - Example: population density → area-weighted average
  - **Extensive observations** scale with the size of the region are typically aggregated by summation.
    - Example: population → summation

Spatial Data in Knowledge Graphs

- **SSN/SOSA:** a widely adopted approach for capturing spatiotemporal phenomena in knowledge
- **RDF Data Cube Vocabulary (QB):** a general framework for representing multidimensional data
- **QB4ST:** a specialized geospatial extension of QB is the RDF Data Cube Vocabulary for Spatio-Temporal Components (QB4ST)
- QB and QB4ST do not specify the semantics of scaling operations, such as upscaling and downscaling
- **Grid-based Modeling:** partitioning the Earth's surface into spatial grid cells, allowing observations to be assigned to standardized spatial units.
  - Often employed in knowledge graphs for spatial indexing of observations
  - Examples are Google's S2 or Uber's H3 grid systems

SOGO Ontology



Spatial Maps

```
:CommunityStateMapping rdf:type owl:NamedIndividual , sogo:SpatialMap ;
                       sogo:sourceFeature crisp:Community ;
                       sogo:targetFeature crisp:State ;
                       sogo:mappingQuery """
PREFIX geo: <http://www.opengis.net/ont/geosparql#>
PREFIX crisp: <http://crispy.ai.wu.ac.at/crisp/>

SELECT * WHERE {
    <SOGO_PART> geo:sfWithin/geo:sfWithin <SOGO_WHOLE> .
    <SOGO_WHOLE> a crisp:State .
    <SOGO_PART> a crisp:Community .
} """ .
```

Spatial map between state and its communities

```
:H3CellCommunityMapping rdf:type owl:NamedIndividual ,
                                    sogo:SpatialMap ;
                        sogo:sourceFeature crisp:Community ;
                        sogo:targetFeature crisp:H3Cell ;
                        sogo:mappingQuery """
PREFIX geo: <http://www.opengis.net/ont/geosparql#>
PREFIX crisp: <http://crispy.ai.wu.ac.at/crisp/>

SELECT * WHERE {
    <SOGO_WHOLE> geo:sfContains <SOGO_PART> .
    <SOGO_WHOLE> a crisp:Community .
    <SOGO_PART> a crisp:H3Cell .
} """ .
```

Spatial map between community and its overlapping H3 cells

Upscaling Example

Use Case: Given population data for individual communities, find the total population of the state.

```
###  Upscaling Procedure
:populationUpscaling rdf:type owl:NamedIndividual, sogo:ScalingProcedure ;
    sogo:hasScalingProperty crisp:population ;
    sogo:hasSpatialMap crisp:CommunityStateMapping ;
    sogo:hasTemporalExtent [ rdf:type time:Year ;
                             time:year "2011"^^xsd:gYear
                           ] ;
    sogo:usesScalingMethod sogo:SumAggregation ;
    rdfs:comment "A scaling procedure for calculating population of states based on community
    population data."@en ;
    rdfs:label "Population Upscaling Procedure"@en .
```

Scaling property

Spatial Map

Allowed periods

Scaling method

Downscaling Example

Use Case: Given population data for a community, find the total population of its overlapping H3 cells.

```
###  Downscaling Procedure
:populationDownscaling rdf:type owl:NamedIndividual , sogo:ScalingProcedure ;
    sogo:hasAuxiliaryProperty crisp:roadLength ;
    sogo:hasScalingProperty crisp:population ;
    sogo:hasSpatialMap crisp:H3CellCommunityMapping ;
    sogo:hasTemporalExtent [ rdf:type time:Instant ;
                             time:inXSDDateTimeStamp "2011-10-31T00:00:00Z"^^xsd:dateTimeStamp
                             ] ;
    sogo:usesScalingMethod sogo:WeightedDisaggregation ;
    rdfs:comment "A scaling procedure for estimating H3 cell populations from community-level
    populations, weighted proportionally by the length of roads within each cell."@en ;
    rdfs:label "Population Downscaling Procedure"@en .
```

Auxiliary property

downscaling metho

SOGO Scaling Implementation

- Goal is Finding value of property ***P*** for region ***R***
- **Step 1:** Identify the defined **Scaling Procedure** instances that specify ***P*** as their scaling property.
- **Step 2:** Identify the procedure that reference a **Spatial Map** for which ***R*** is defined as the target feature.
- **Step 3:** Using the selected Spatial Map, retrieve the set of source regions involved in the scaling process, together with the required properties the given *temporal extent*
- **Step 4:** Apply the scaling method specified by the Scaling Procedure to the retrieved property values and calculate the corresponding scaled value.
- **Step 5:** Use the resulting value to construct a synthetic ***P*** observation for region ***R***.

Generated Observation

```
<id/observation/bf2adddd553646fd95e74047536b5d91> a sosa:Observation ;
        sosa:resultTime "2011-10-31T00:00:00Z"^^xsd:dateTime ;
        sosa:hasFeatureOfInterest <http://crispy.ai.wu.ac.at/crisp/id/state/1> ;
        sosa:observedProperty <http://crispy.ai.wu.ac.at/crisp/population> ;
        sosa:hasSimpleResult "285685"^^xsd:integer ;
        sosa:usedProcedure <http://crispy.ai.wu.ac.at/crisp/populationUpscaling> .
```

Calculated value

preserving the provenance of the derived value

Conclusions & Future Work

- Using SOGO ontology we can capture spatial maps and encode the scaling logic of observations within knowledge graphs
- It allows data owners and data publishers, who possess deep contextual knowledge, to explicitly encode scaling logic alongside the data itself
- The implemented use cases illustrate how spatial knowledge graphs can benefit from semantically rich, ontology-driven scaling processes
- It could also enable AI agents to generate new, reliable observations based on existing observations
- In future, we plan to extend our approach to more complex scaling methods such as interpolation and extrapolation



GitHub repository including SOGO ontology and experimental implementation with a sample dataset

https://github.com/anjomshoaa/sogo

Thanks for your attention!

Amin Anjomshoaa
https://anjomshoaa.github.io/
amin.anjomshoaa@wu.ac.at